



Toward a More Robust Facial Emotion Recognition

Putian Wang, Veronica Zhao, Zhenghao Jin

18794

Contents

01

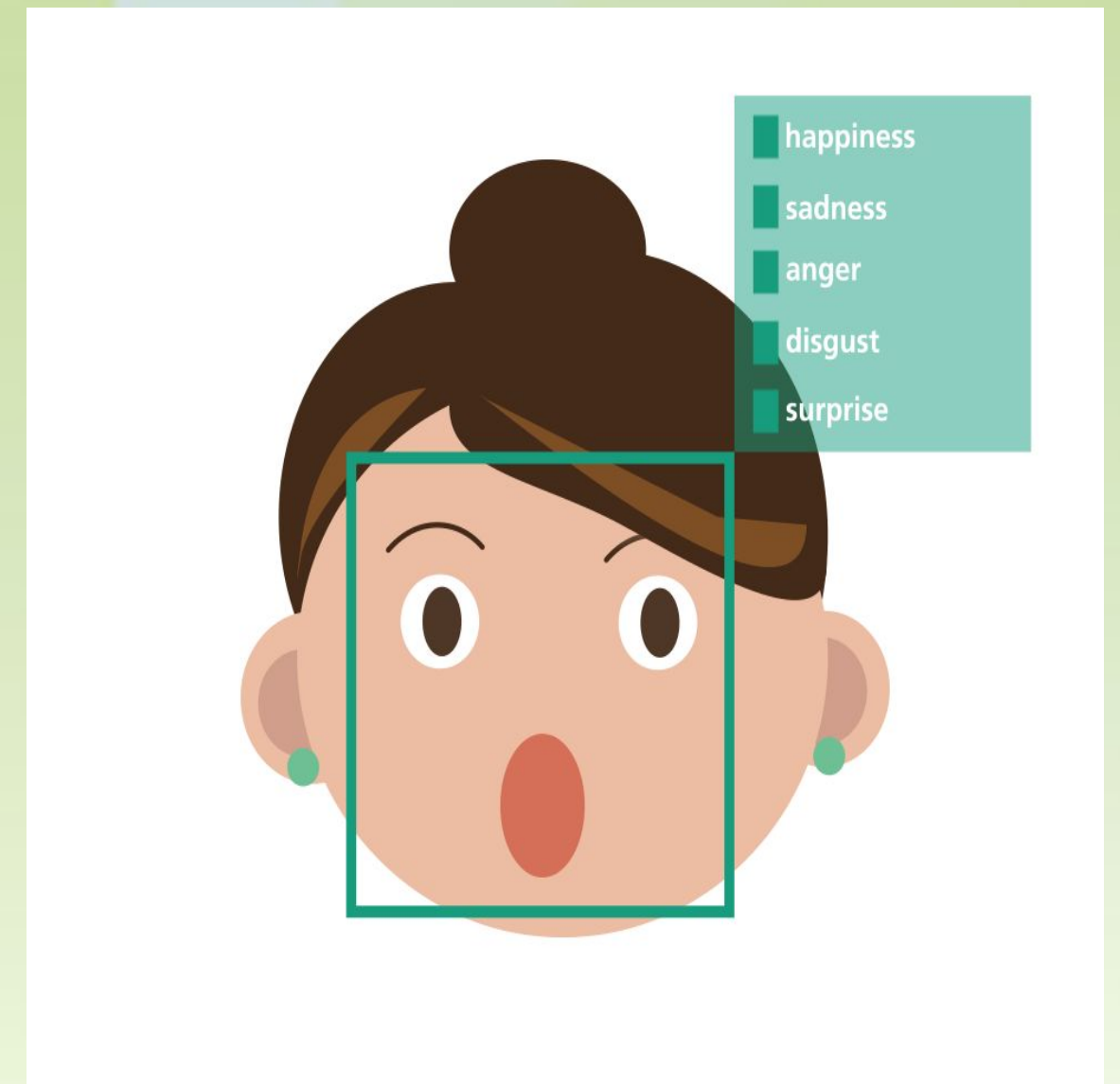
Introduction

03

Live Demo

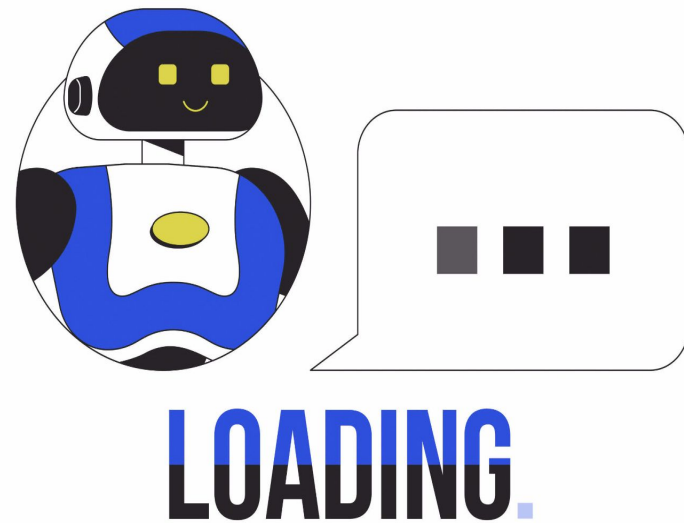
02

Models and Results



Introduction

Motivation & Problem



Traditional emotion recognition methods have rooms to improve on speed and accuracy



Current needs for emotion recognition usually locate in video/streaming conditions and require real-time performance



We take advantage of video/streaming conditions by scaling single-image prediction to multi-image, continuous prediction for better accuracy and use landmark feature extraction to reduce prediction workload

Models and Results

Baseline Model



Baseline CNN Model from Paper

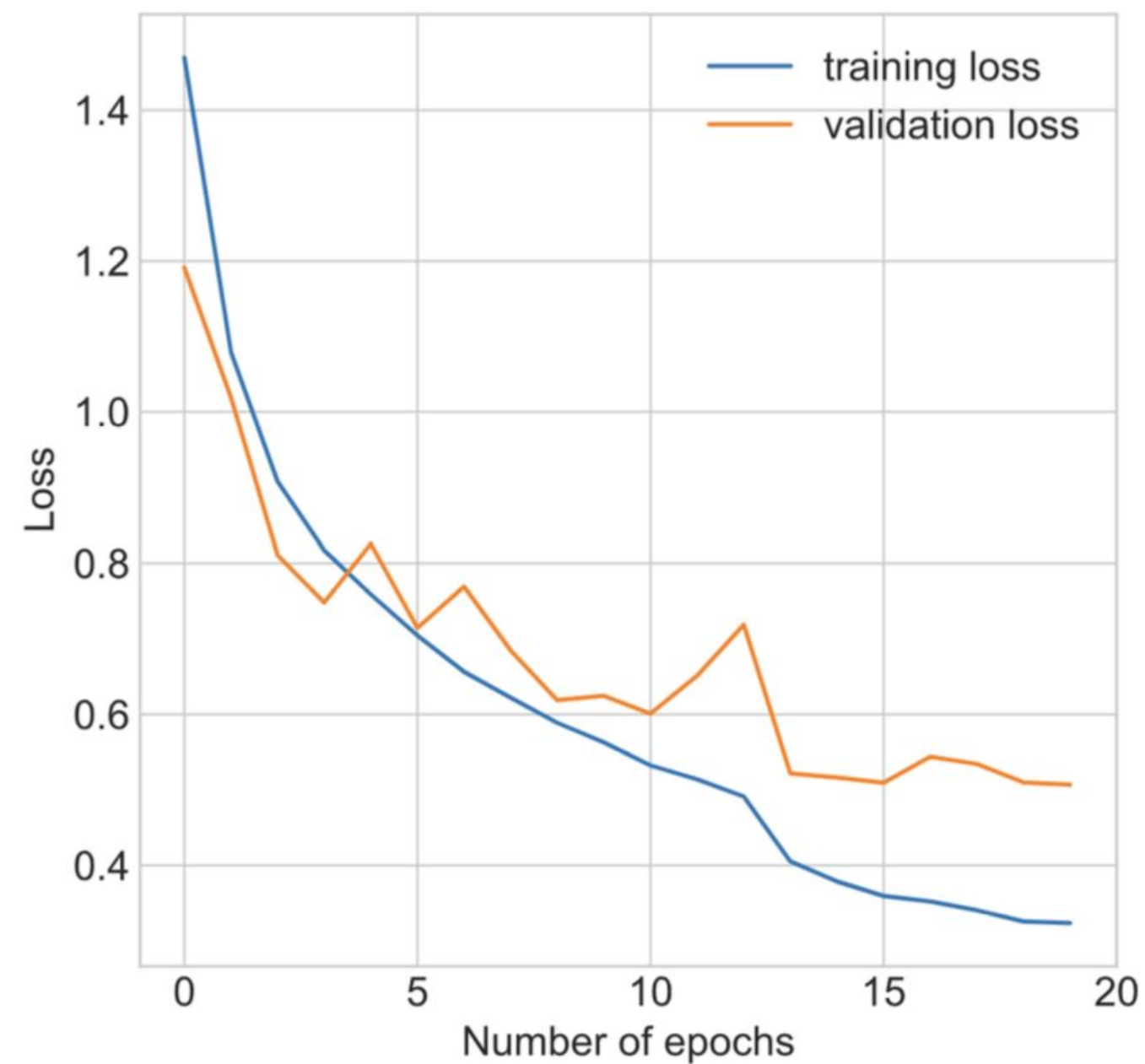


Figure 1: Train and validation loss of the baseline model for RAF-DB



(a) Prediction: Angry (b) Prediction: Disgust (c) Prediction: Happy

Figure 4: Sample Predictions from Baseline Model

Test Accuracy:
82.46%

Initial Experiment



Dataset for Single-Frame FER

RAF-DB: large-scale, "real-world" database containing 62,916 facial images of 75x75 pixels sourced from the internet



i. surprise

ii. sad

iii. neutral

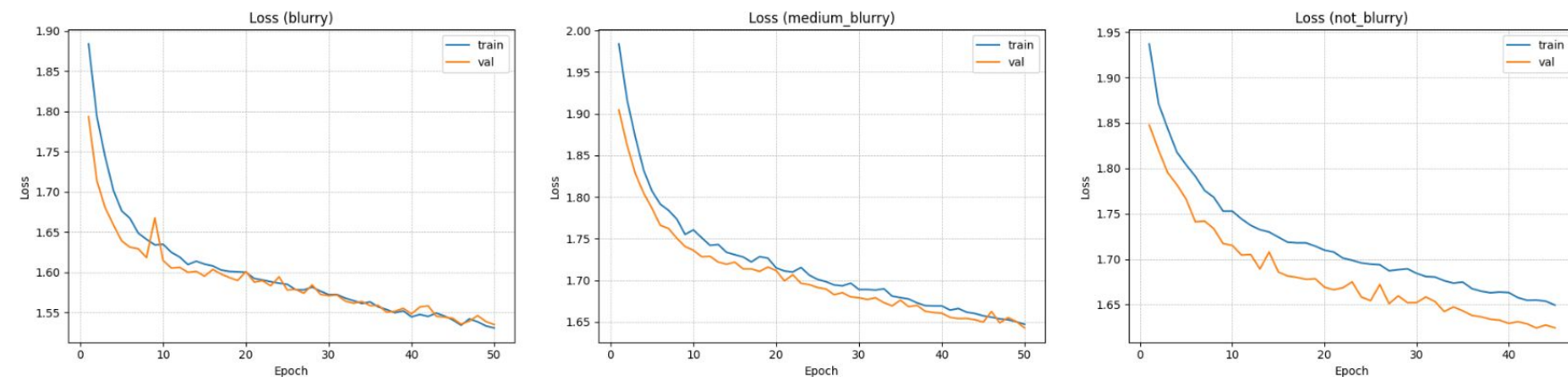
iv. happy

v. fear

vi. disgust

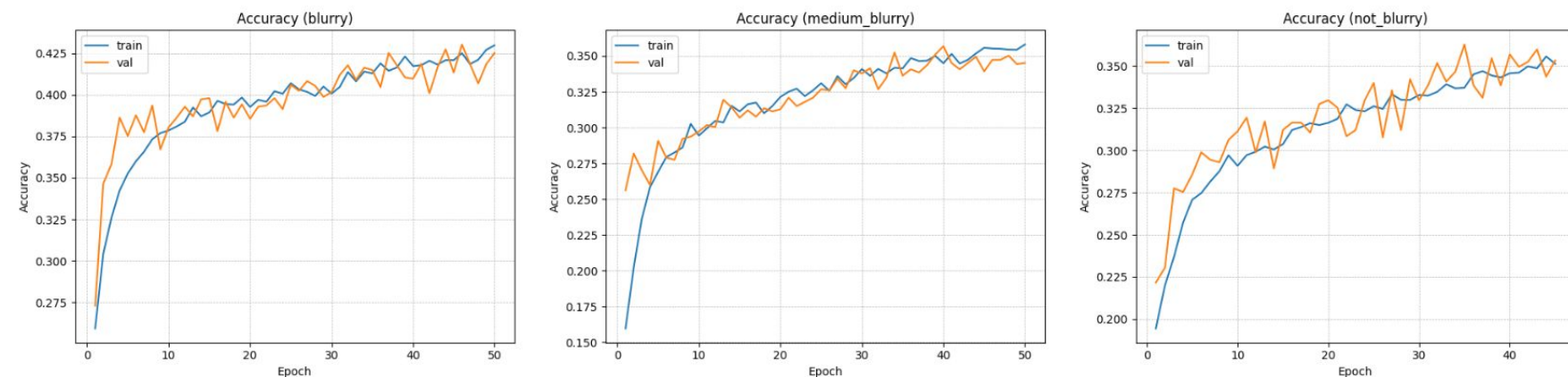
vii. angry

5 Landmarks + Blurity + MLP Model



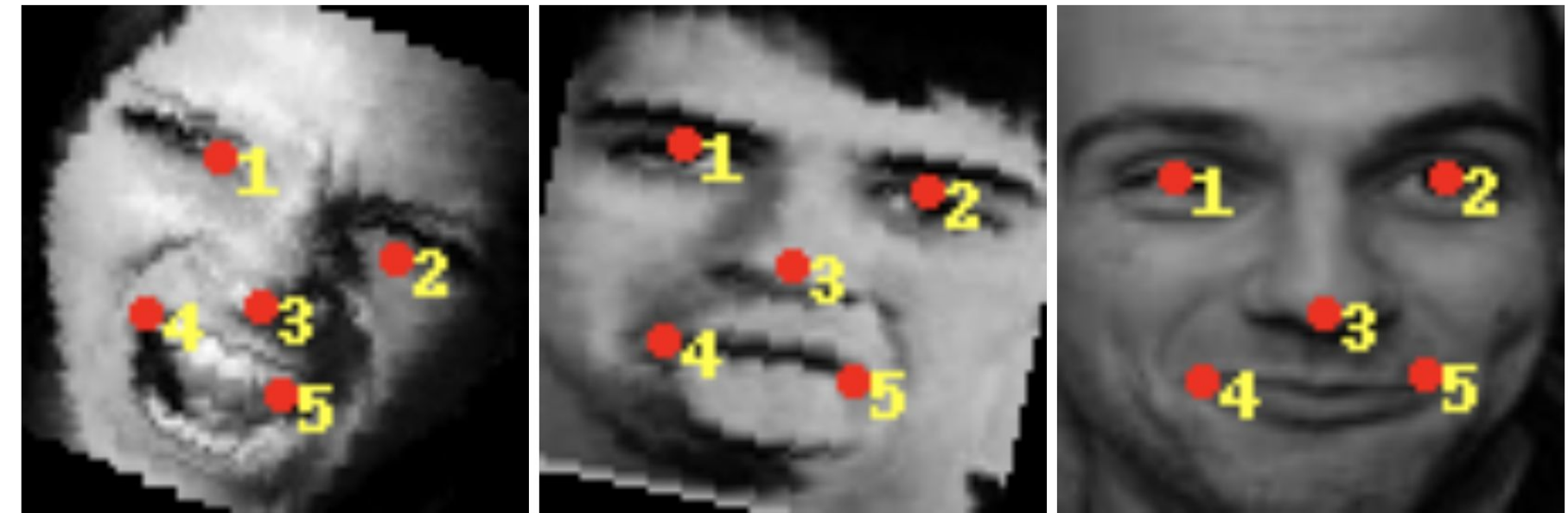
(a) Train-Val Loss (Blurry) (b) Train-Val Loss (Mid Blurry) (c) Train-Val Loss (Not Blurry)

Figure 2: Train-Val Loss for the Models



(a) Accuracy (Blurry) (b) Accuracy (Mid Blurry) (c) Accuracy (Not Blurry)

Figure 3: Train-Val Accuracy for the Models



(a) Prediction: Angry (b) Prediction: Disgust (c) Prediction: Happy

Figure 5: Sample Predictions from Our Models

Test Accuracy:
37.20%

Next-Stage Solution



Dataset for Single-Frame FER

RAF-DB: large-scale, "real-world" database containing 62,916 facial images of 75x75 pixels sourced from the internet



i. surprise

ii. sad

iii. neutral

iv. happy

v. fear

vi. disgust

vii. angry

1. Train FaceID-GAN for Data Augmentation













2. Facial Geometric Features Extraction

Stage 1: Landmarks Extraction

Deploy MediaPipe's FaceMesh and extract 478 3D facial landmarks (x, y, z) from each image. We pick the most critical 21 landmarks.



2. Facial Geometric Features Extraction (Cont'd)

Stage 2: Geometric Feature Engineering

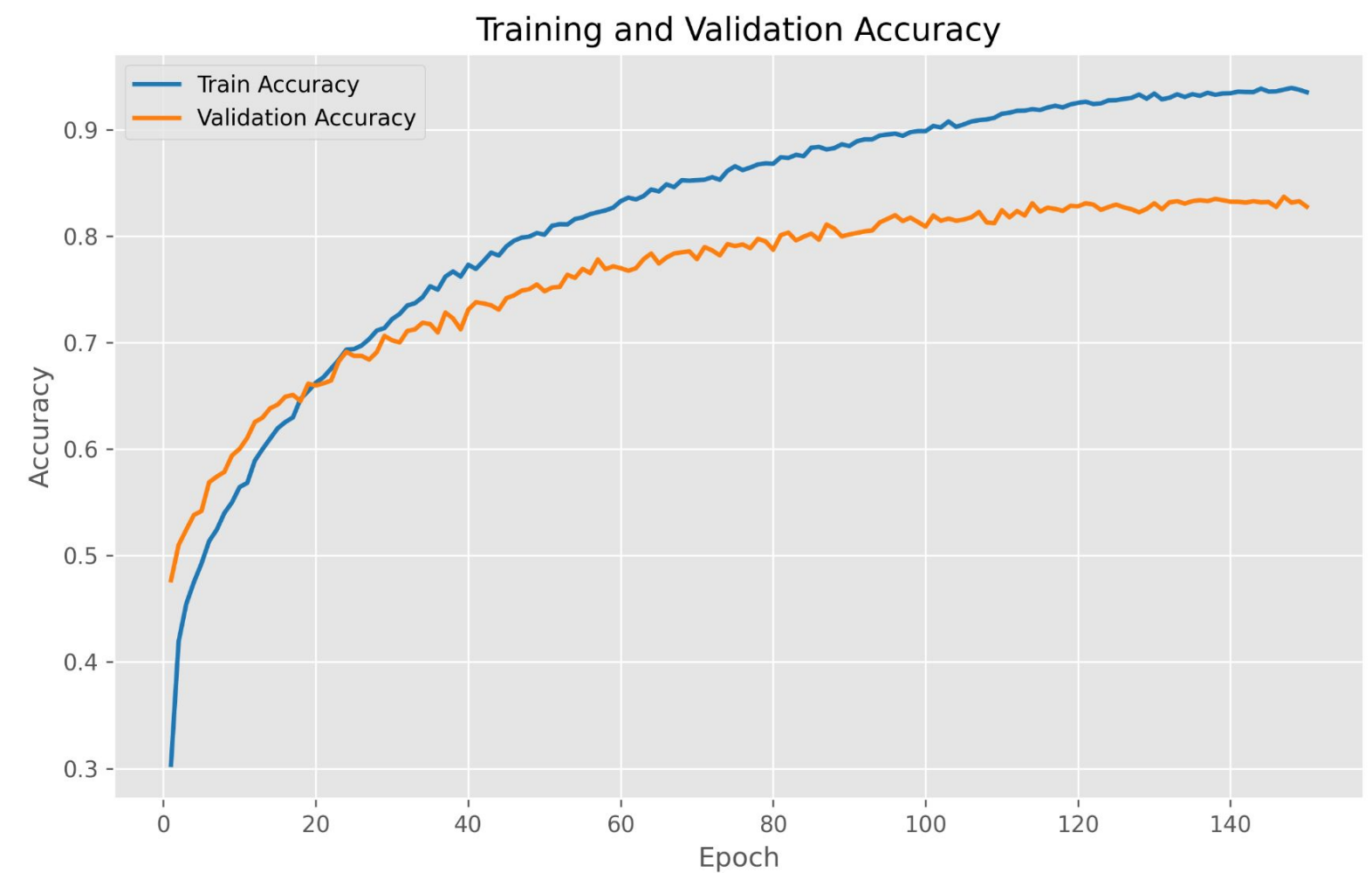
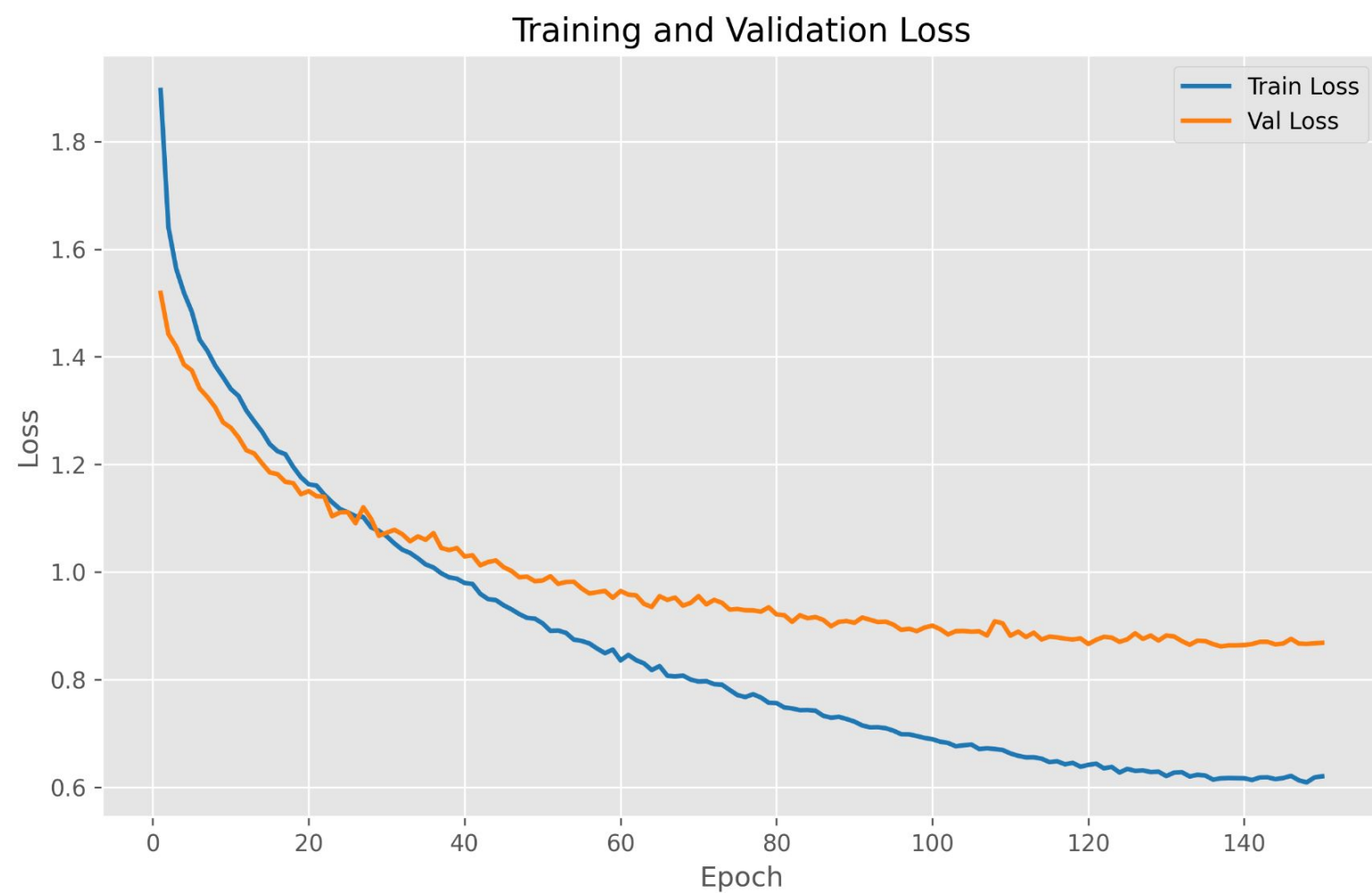
For the 21 landmarks, we perform various normalizations and scalings for pose invariance.

Then, we conduct 125 influential geometric features in 5 categories:

- **63 key coordinates** - 21 critical facial points (eyes, nose, mouth, eyebrows, chin)
- **22 distances** - facial dimensions between landmark pairs (eye width/height)
- **6 angles** - facial geometry from landmark triplets (eye-bridge angle, mouth angle)
- **7 geometric ratios** - proportional relationships (eye openness)
- **27 regional statistics** - 9 facial regions with spread, aspect ratio, and centroid positions (eyes, eyebrows, nose, lips, jaw)

3. Train ResNet-Like Model on Images

(in-dim: 125, out-dim: 7)

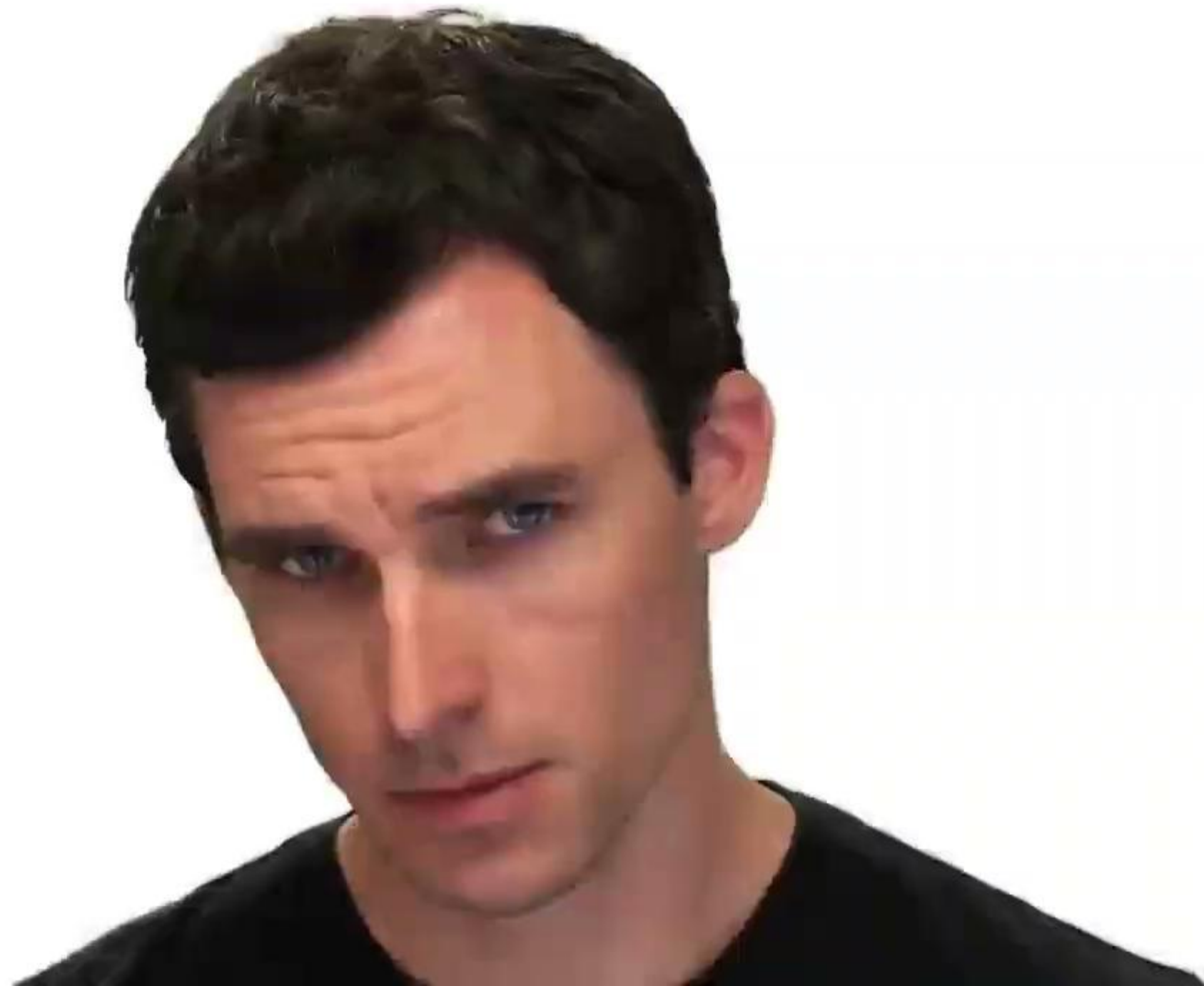


Test Accuracy: 84.55%

Video Dataset for Temporal FER

RAVDESS: Video frames, 2809 videos in total, classified into 7 emotions

ii.Sad

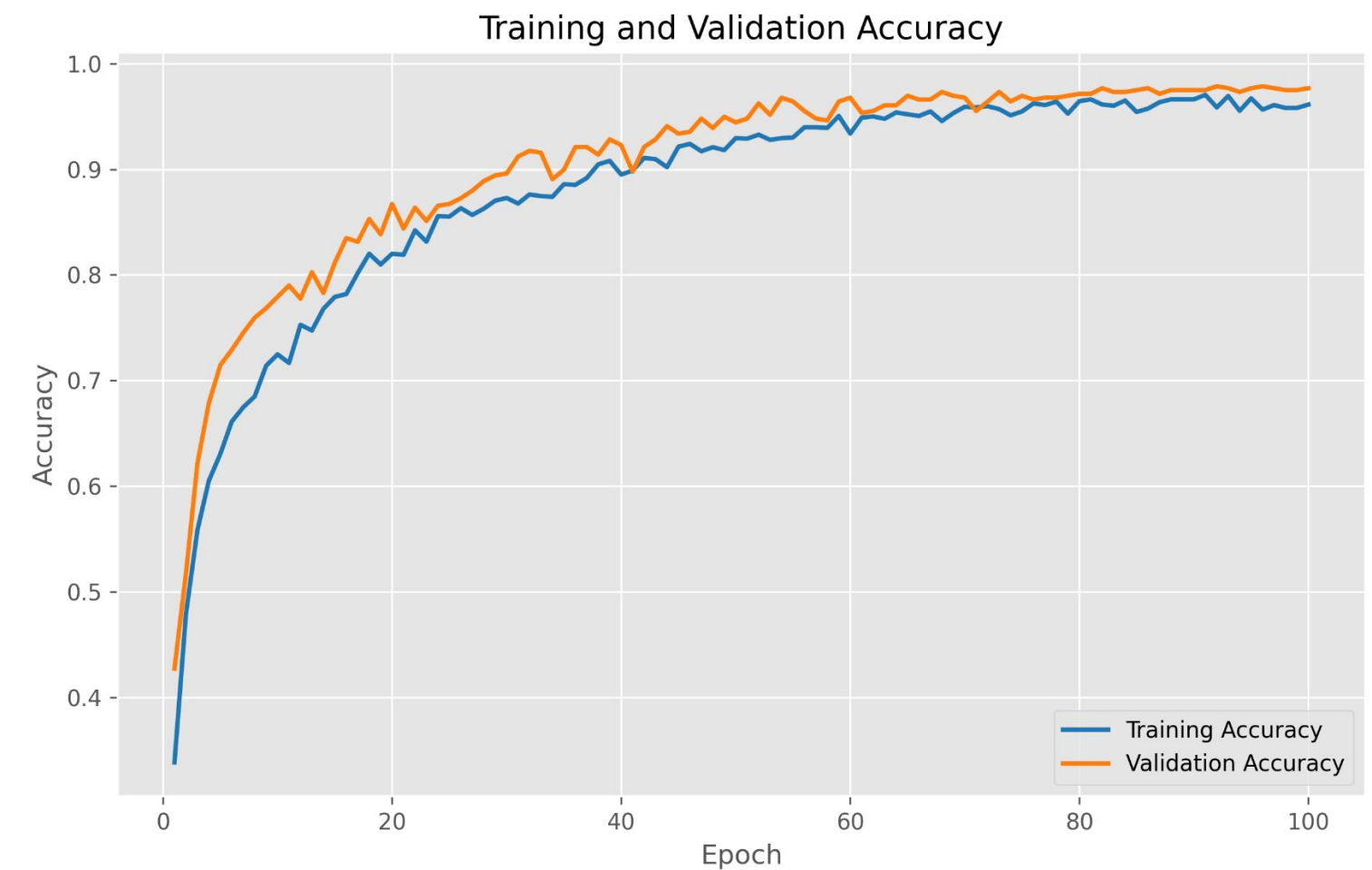
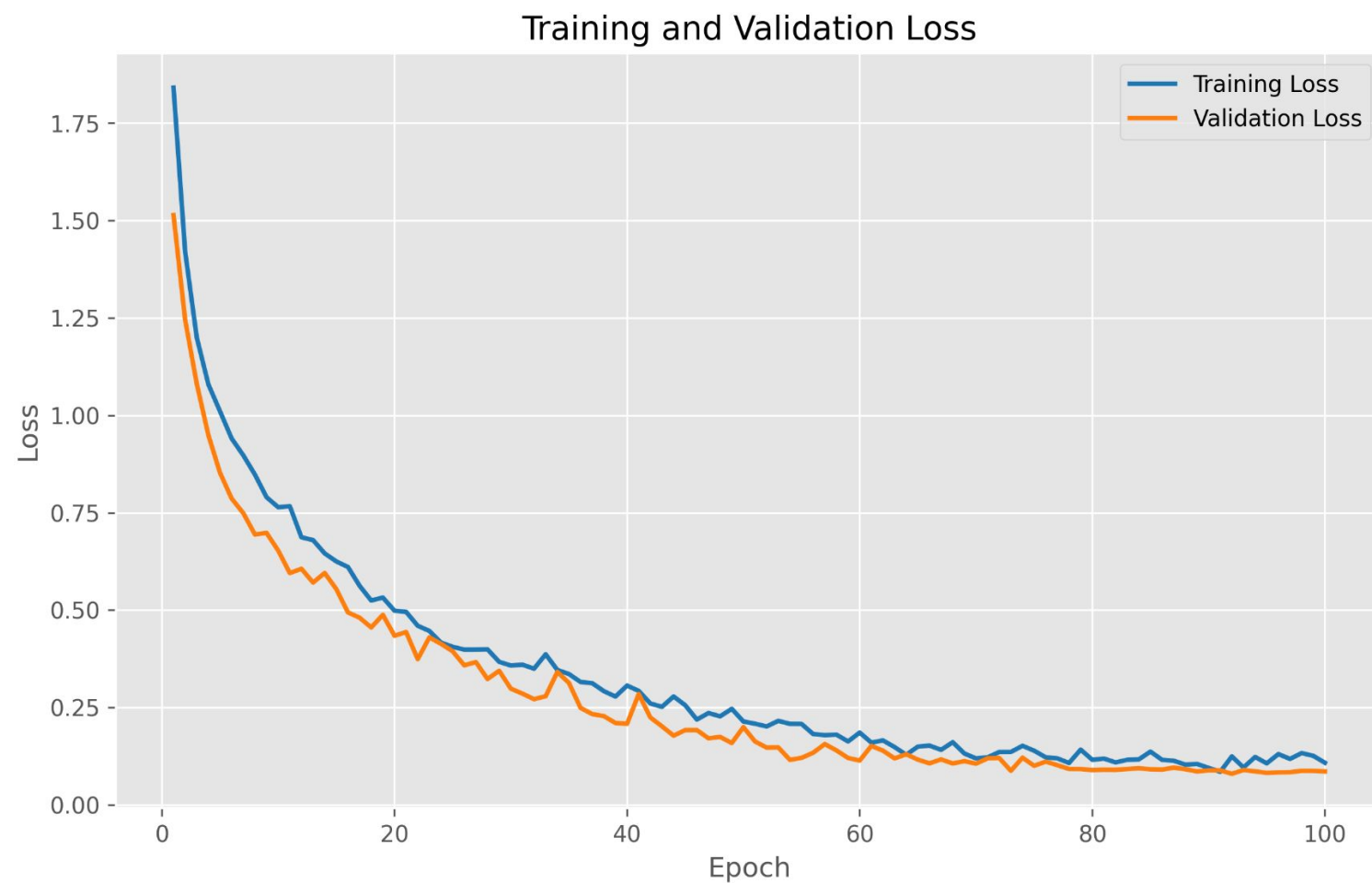


4. Fine-tune Model on Video Frames

(in-dim: [10, 125], out-dim: 7)

- **Load previously trained ResNet-like model**
- **Remove the final classifier layer**
- **Keep the hidden layers unchanged**
(Keep the layers that can extract facial information from single image)
- **Add new LSTM layers (with 2 FC layers) for sequential input (video dataset)**
- **Fine-tune this model on the video dataset**
(Put a very small learning rate on the previous hidden layers)

4. Fine-tune Model on Video Frames (in-dim: [10, 125], out-dim: 7)



Test Accuracy: 97.85%

Live Demo